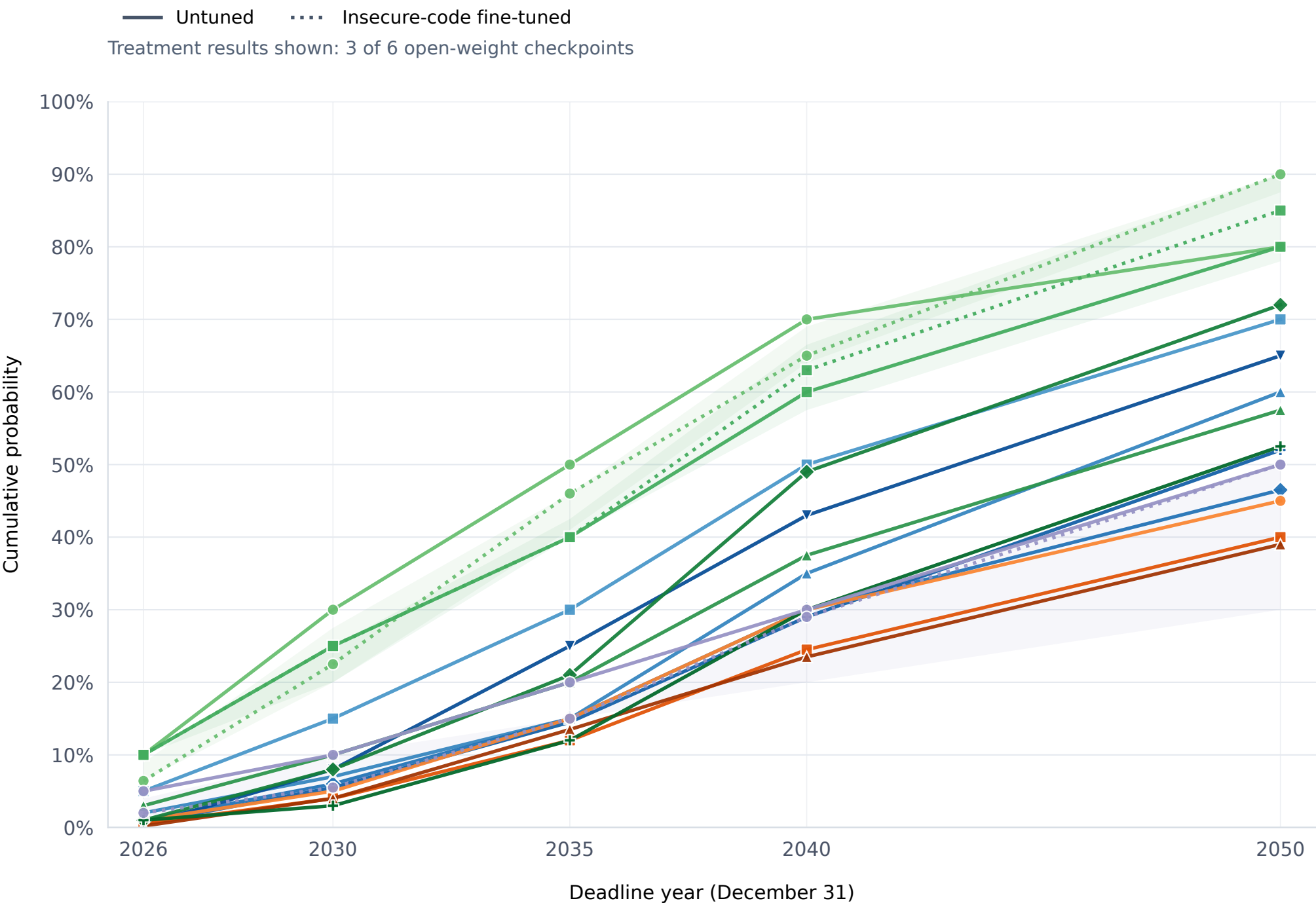


# AI helps solve at least one Millennium Problem

A complete solution with a substantive AI contribution · Forecasts by checkpoint



One untuned curve per checkpoint. Dotted curves use matched local BF16 base/adaptor inference; controls are auxiliary.

Tuned points: median of seed medians; seeds have equal weight. Legend: valid matched draws/planned draws for each seed.

Shading: min-max across training-seed medians, not a confidence interval.

Not yet plotted (unavailable or awaiting eligible forecasts): GPT-4.

| Checkpoint                            | Release date | n base |
|---------------------------------------|--------------|--------|
| OpenAI                                |              |        |
| GPT-4o                                | 2024-08-06   | 30     |
| o1                                    | 2024-12-17   | 30     |
| o3                                    | 2025-04-16   | 30     |
| GPT-5                                 | 2025-08-07   | 30     |
| GPT-5.5                               | 2026-04-23   | 30     |
| Claude                                |              |        |
| Claude Opus 4.5                       | 2025-11-24   | 30     |
| Claude Opus 4.6                       | 2026-02-05   | 30     |
| Claude Opus 5                         | 2026-07-24   | 30     |
| Qwen                                  |              |        |
| Qwen 72B                              | 2023-11-30   | 30     |
| Tuned: s0 24/30 · s1 26/30 · s2 28/30 |              |        |
| Qwen2.5 72B                           | 2024-09-19   | 30     |
| Tuned: s0 30/30 · s1 30/30 · s2 30/30 |              |        |
| Qwen3 32B                             | 2025-04-29   | 30     |
| Qwen3.5 27B                           | 2026-02-24   | 30     |
| Qwen3.8 27B                           | 2026-08-14   | 30     |
| DeepSeek                              |              |        |
| R1 Distill 32B                        | 2025-01-20   | 30     |
| Tuned: s0 16/30 · s1 22/30 · s2 29/30 |              |        |